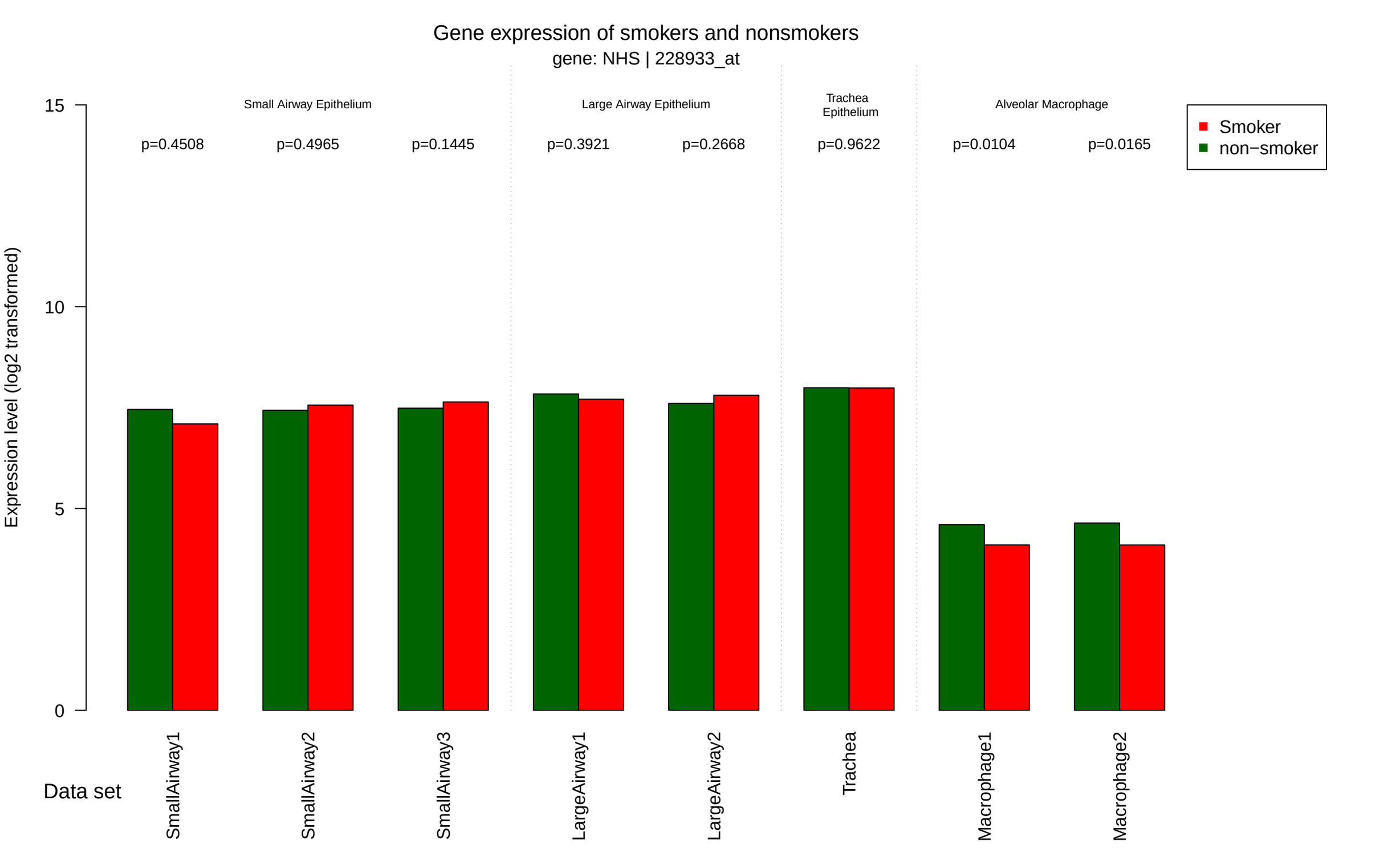
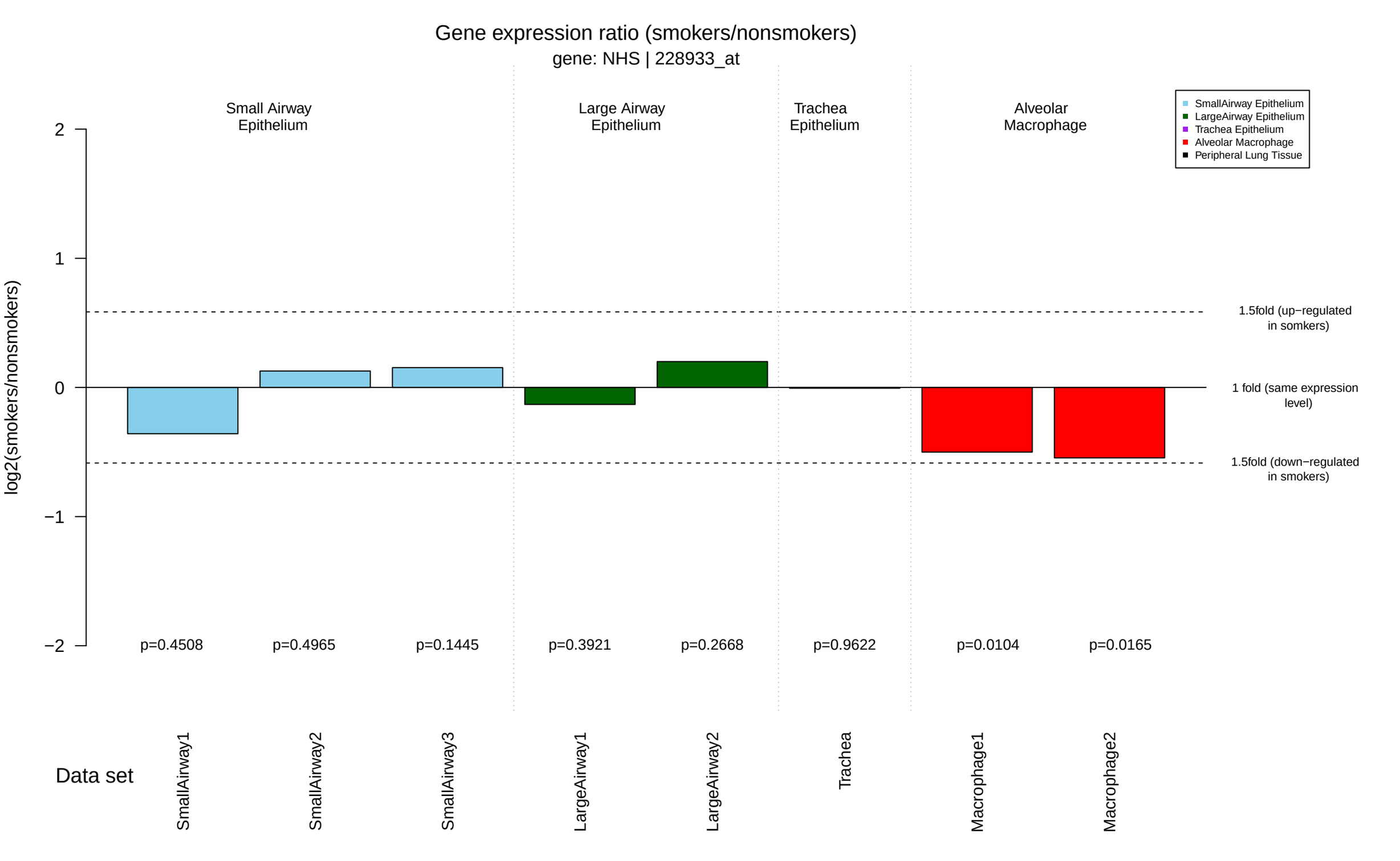
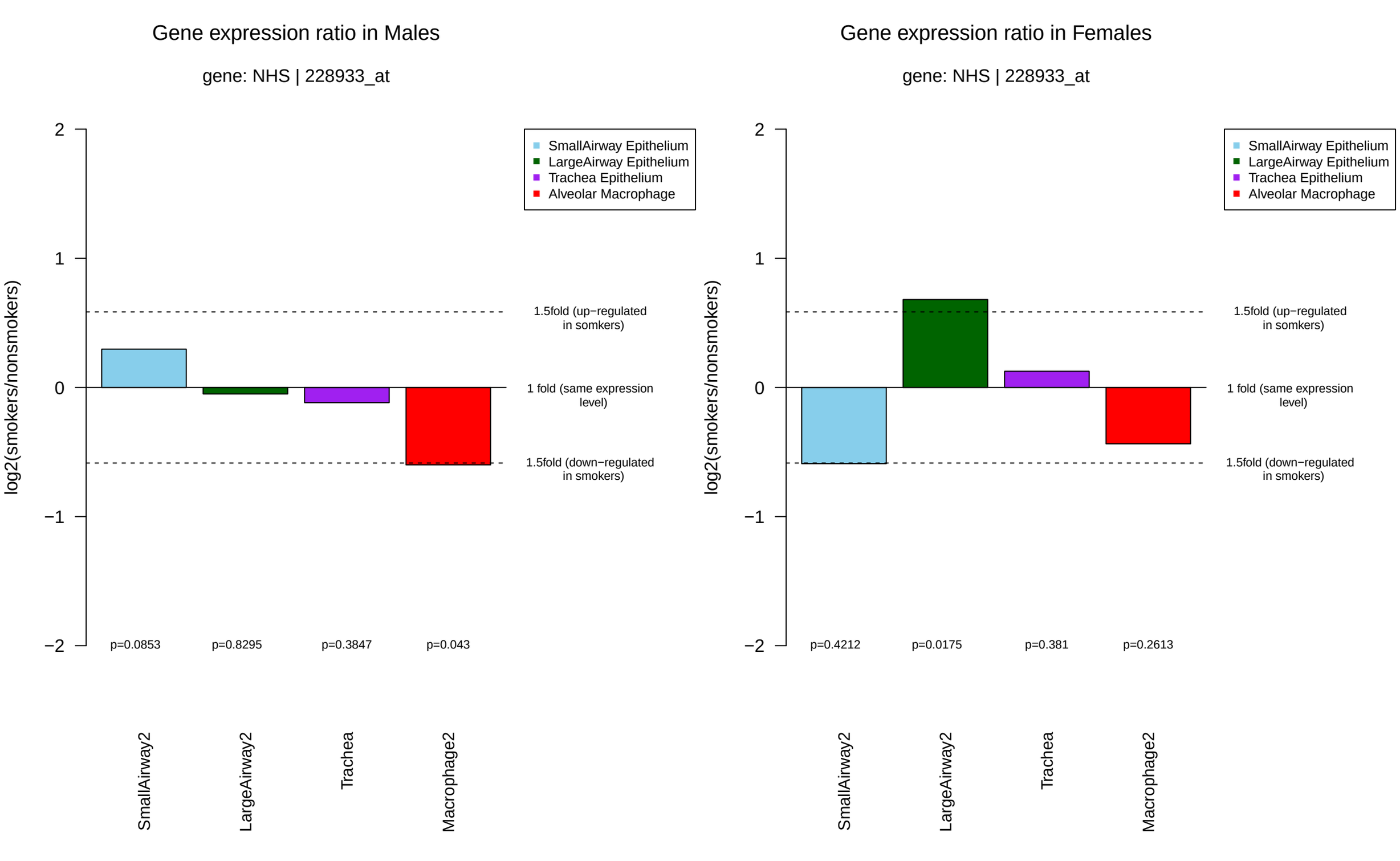
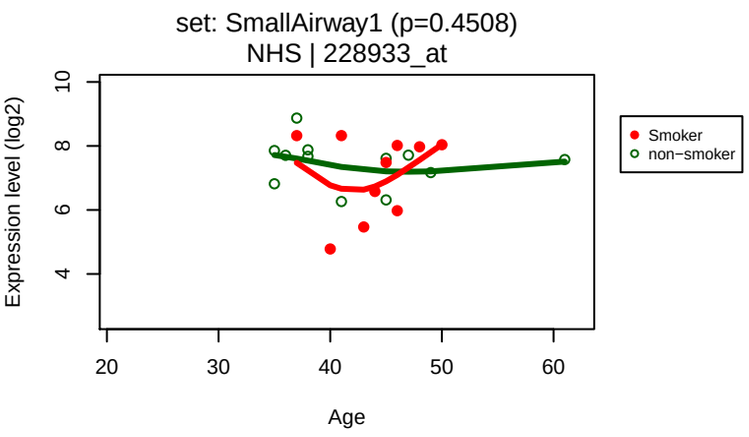
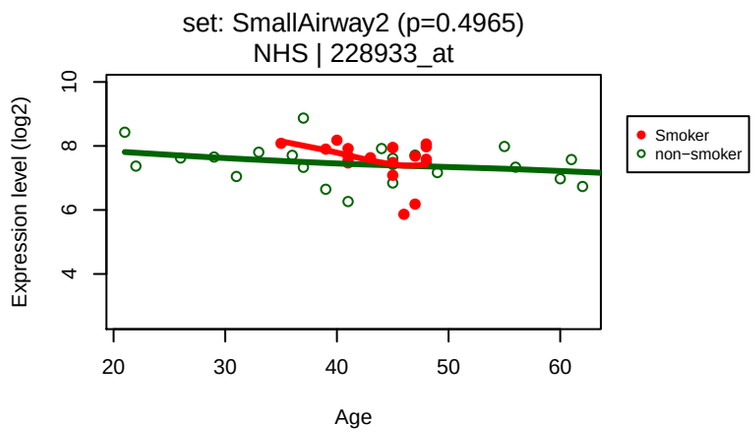
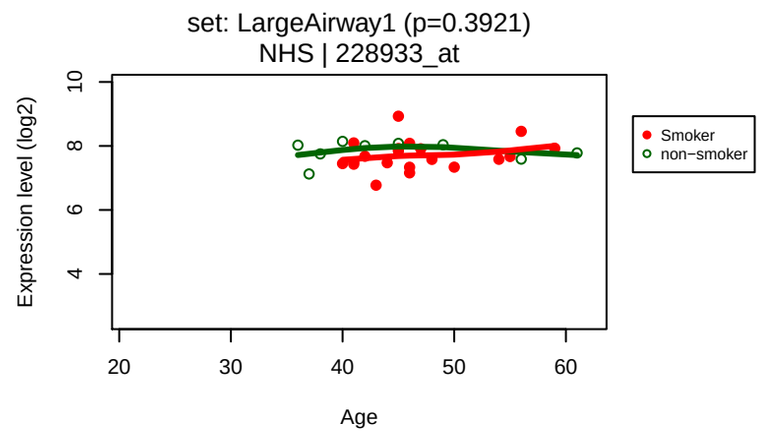

Gene expression of smokers and nonsmokers
gene: NHS | 228933_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4508
p=0.4965
p=0.1445
p=0.3921
p=0.2668
p=0.9622
p=0.0104
p=0.0165
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

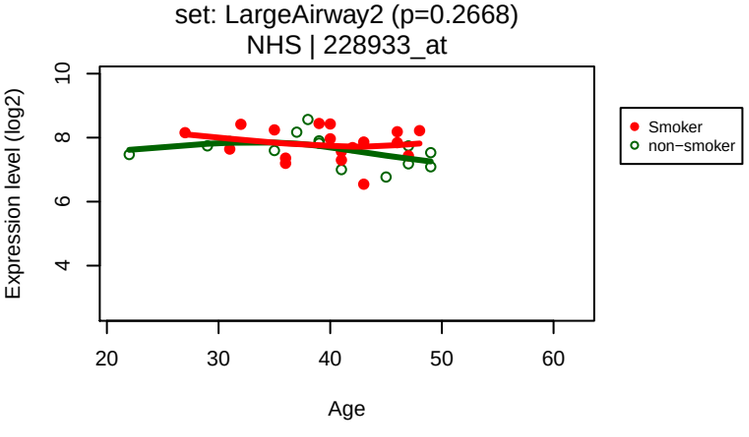

Gene expression ratio (smokers/nonsmokers)
gene: NHS | 228933_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4508
p=0.4965
p=0.1445
p=0.3921
p=0.2668
p=0.9622
p=0.0104
p=0.0165
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

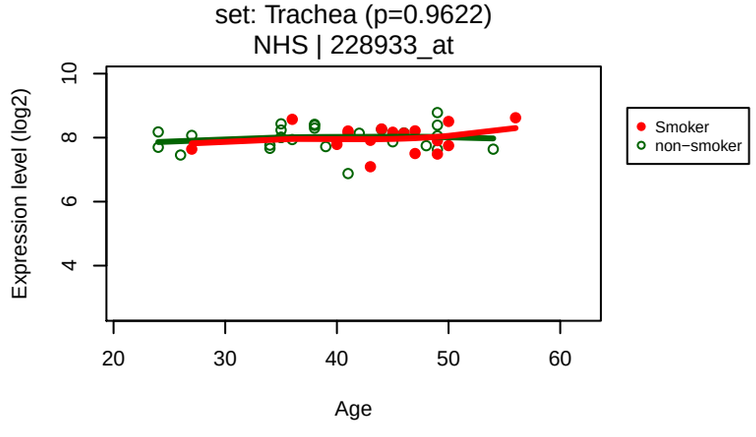

Gene expression ratio in Males
gene: NHS | 228933_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0853
p=0.8295
p=0.3847
p=0.043
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NHS | 228933_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4212
p=0.0175
p=0.381
p=0.2613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

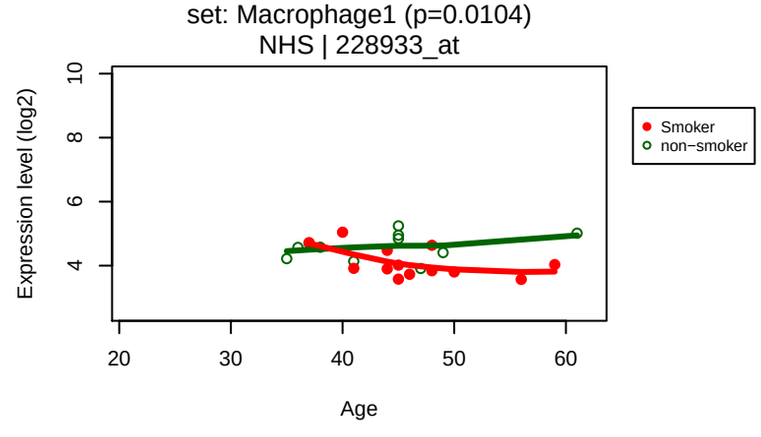

set: SmallAirway1 (p=0.4508)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker

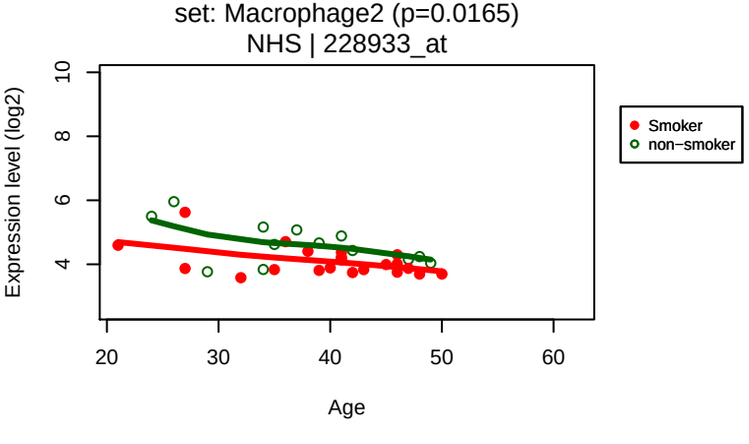

set: SmallAirway2 (p=0.4965)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker

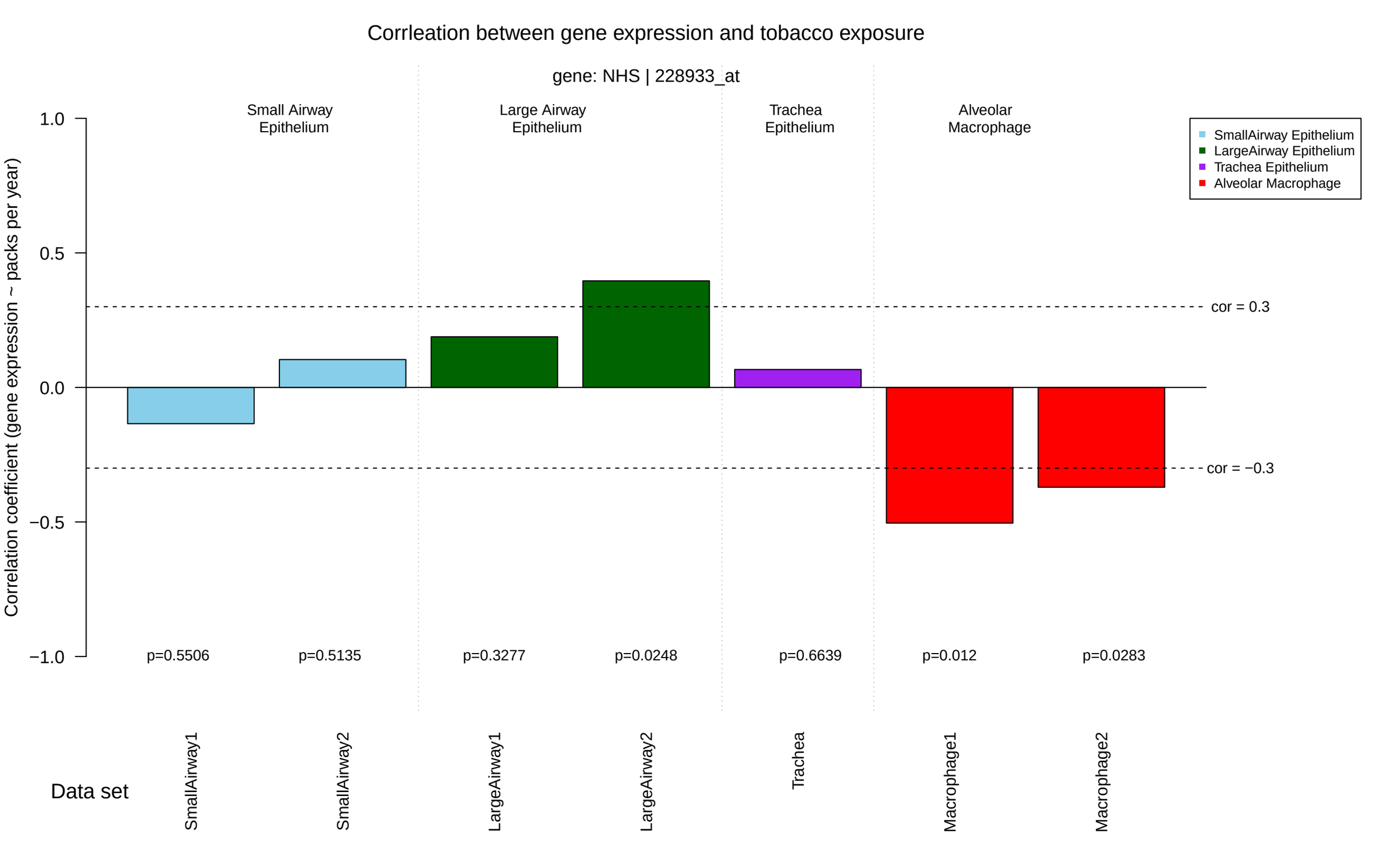

set: LargeAirway1 (p=0.3921)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2668)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9622)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0104)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0165)
NHS | 228933_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NHS | 228933_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5506
p=0.5135
p=0.3277
p=0.0248
p=0.6639
p=0.012
p=0.0283
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2